

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Efficient posterior approximation remains a central and technically demanding problem in Bayesian inference. Stein Variational Gradient Descent (SVGD), a deterministic variational inference procedure that transports an ensemble of particles toward a target law, derives its dynamics from a reproducing kernel Hilbert space (RKHS) and is consequently highly sensitive to the selected kernel, whose form directly shapes both convergence behavior and approximation fidelity. Although the widely used median heuristic supplies a convenient bandwidth prescription, its limited flexibility and empirically observed deterioration, particularly in high-dimensional regimes, motivate a more principled adaptive alternative. In this work, we develop a strategy for selecting kernel parameters over an abstract parametric kernel family during inference. Recent convergence analyses formulated through the kernelized Stein discrepancy (KSD) indicate that performance can be improved by choosing parameters that maximize the KSD; building on this observation, we introduce Adaptive SVGD (Ad-SVGD), an alternating scheme in which particle transport by SVGD is interleaved with gradient-ascent tuning of kernel bandwidths. We give a simplified theoretical analysis that extends fixed-kernel KSD convergence arguments to the adaptive setting and establishes convergence properties for the maximal KSD over the prescribed kernel class. The accompanying experiments support this mechanism empirically, showing that Ad-SVGD consistently improves upon standard heuristic choices across a range of tasks.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle-based methodology for approximate Bayesian inference that has become a prominent alternative to traditional Markov Chain Monte Carlo (MCMC) procedures. The method evolves an ensemble of particles along update directions obtained from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, while constraining those directions to the unit ball of a reproducing kernel Hilbert space (RKHS). A persistent limitation of this construction is its pronounced dependence on kernel choice, which materially affects algorithmic performance [8, 28]; moreover, the resulting particle approximation frequently underestimates posterior variance [2]. These phenomena have contributed to the common view that SVGD, in its original form, deteriorates as the dimension of the underlying state space grows. We revisit this assessment by introducing an adaptive kernel-parameter selection mechanism that dynamically tunes the kernel during inference through maximization of the kernelized Stein discrepancy (KSD), thereby enabling more effective particle transport in complex and high-dimensional settings.

1.1 Related work.

Since its introduction [18], SVGD has been adopted as a broadly applicable tool for approximate Bayesian inference across diverse machine-learning settings [22, 24, 30, 15]. Parallel theoretical developments have clarified several aspects of its behavior: mean-field convergence has been investigated in both continuous-time [23, 8, 5] and discrete-time [16, 32] formulations, while finite-particle convergence rates have been established under various structural assumptions [3, 33]. In addition, SVGD dynamics have been interpreted as gradient flows on spaces of probability distributions [19, 8], in analogy with the variational perspective of Jordan, Kinderlehrer, and Otto [14].

The efficacy of SVGD depends crucially on the kernel function, because the kernel determines inter-particle interactions and thereby influences the convergence profile of the overall method (see also Figures 1 and 9). Existing convergence statements are often formulated at the mean-field level in terms of KSD, yet the implication of KSD convergence for weak convergence itself depends on the selected kernel [11]. The median heuristic [12], although computationally simple and widely used, lacks a general theoretical justification and is known to deteriorate as problem dimensionality increases [2, 40, 36]. Complementary work has therefore sought to mitigate high-dimensional degradation through modified algorithms [7, 37, 10, 21], and related variants have used neural networks to learn the update direction itself [17, 39].

The closest antecedent to our contribution is the mixture-of-kernels framework of Ai et al. [1], which defines a KSD associated with a weighted linear combination of kernels and learns the corresponding weights by maximizing this multiple-kernel discrepancy. That construction, however, is restricted to finite kernel bases and does not address the continuous kernel-parameter optimization considered here.

1.2 Contributions.

This work asks whether SVGD, without architectural augmentation or substantial algorithmic redesign, can achieve competitive performance when equipped with a principled and adaptively selected kernel. By isolating kernel selection from other modifications appearing in SVGD variants, we seek to clarify the extent to which kernel choice alone governs the effectiveness of the original method and how far adaptive kernel learning can extend its practical capabilities.

Our principal contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We propose a method that updates kernel parameters dynamically by maximizing the KSD during SVGD inference. Unlike the median heuristic, which typically supplies a single scalar bandwidth, our approach permits the simultaneous optimization of multiple continuous kernel parameters and therefore provides substantially greater flexibility during the particle updates.
- (ii) **Theoretical Analysis.** We substantiate the method by analyzing it in the discrete-time mean-field setting, extending existing fixed-kernel convergence results for SVGD. In particular, we show that the supremum of the KSD over a parameterized kernel class converges to zero as the particle law approaches the target, and, under a Stein logarithmic Sobolev inequality, we further derive mean-field iteration-complexity bounds.
- (iii) **Empirical Validation.** Through numerical experiments, we demonstrate that adaptive kernel selection systematically improves upon the median heuristic and mitigates the variance collapse observed in several SVGD approximations.

2 Mathematical background

We begin with a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and its associated RKHS $\mathcal{H}_0$. Let $\mathcal{H}$ denote the d -fold Cartesian product of $\mathcal{H}_0$, equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$, and let $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$ be the canonical feature map. Throughout, $x \cdot y$ denotes the Euclidean inner product and $\nabla \cdot$ the divergence operator. We write $\mathcal{P}(\mathbb{R}^d)$ for the space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, and $\mathcal{P}_p(\mathbb{R}^d)$ for the subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is defined by

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify discrepancies between probability measures through the largest deviation of expectations over a prescribed class of test functions. When the target distribution is intractable, Stein's method [35] provides a mechanism for constructing test functions with zero target expectation, thereby making such comparisons operational. Specifically, for the operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. The resulting *kernelized Stein discrepancy* (KSD) is defined as

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. This optimization problem is attained by $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, and Φ_k is the feature map associated with the kernel k . Consequently, the supremum reduces to the RKHS norm of ψ , yielding the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and an initial reference distribution μ_0 , SVGD transports μ_0 toward an approximation of π by setting $\mu_{n+1} := T_{\#} \mu_n$, where $T_{\#} \cdot$ denotes push-forward under the map $T(x) = x + \gamma \psi^{\mu_n}(x)$, and where the vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ represents the steepest-descent direction. The justification is that the optimizer of Equation (1) implies that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes $\frac{d}{d\gamma} \text{KL}(T_{\#} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], with $\text{KL}(\cdot \| \cdot)$ denoting KL divergence. In particular, this choice yields

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\#} \mu_n \| \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n | \pi). \quad (2)$$

Iterating the maps T generated in this manner over a particle ensemble $\{X_0^i\}_{i=1}^M$ sampled from μ_0 gives the particle recursion

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because convergence of SVGD in KL divergence is difficult to establish and typically requires restrictive assumptions, recent analyses have increasingly focused on convergence measured by the kernelized Stein discrepancy (KSD) [16, 33, 32]. Nevertheless, KSD minimization alone does not automatically imply weak convergence: the associated mean-field measures may fail to be tight, although tightness is necessary for weak convergence of probability measures on Polish spaces. Moreover, as is already apparent from the variational formulation in Equation (1), the RKHS structure—and hence the kernel itself—determines how effectively convergence in KSD translates into weak convergence [11].

To address this dependence and to strengthen the practical convergence behavior, we formulate an adaptive variant of SVGD over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, we write KSD_θ for the corresponding *kernelized Stein discrepancy*, ψ_θ^μ for the optimal update direction, and Φ_θ for the associated feature map. This explicit parameterization permits kernel selection to be integrated into the optimization process itself, thereby accounting for the kernel’s influence on convergence while avoiding exclusive reliance on a fixed, preselected bandwidth.

The central principle underlying our construction is that, at a given particle configuration, selecting the kernel that maximizes the KSD between the empirical particle law and the target is advantageous. The intuition follows directly from the SVGD descent identity: the instantaneous decrease in KL divergence is proportional to the squared KSD at the current particle measure (see Equations (2) and (7)). Consequently, at each iteration, choosing the kernel with the largest KSD corresponds to maximizing the local rate of KL decrease. This interpretation is also consistent with geometric analyses of SVGD as a gradient flow, which indicate that kernels inducing larger KSD values yield more favorable convergence behavior when the associated RKHSs are compared [28, 8].

Whereas standard SVGD typically seeks KSD convergence for a fixed kernel, our algorithm targets the worst-case KSD over the admissible kernel class. In contrast to the median heuristic used by Liu and Wang [18], the proposed mechanism applies to arbitrary parameterized kernel families: at each step, we seek an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n \mid \pi).$$

Operationally, we approximate this selection by performing one or more gradient-ascent steps on the KSD, using step size $s > 0$, before carrying out the SVGD particle update; to reduce computational cost, we also allow the kernel parameter to remain fixed during selected SVGD iterations (see Algorithm 1). This option is represented by the user-specified decision variable `paramupdate`. The particle-transport component can be replaced by more elaborate SVGD variants, such as adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates, without altering the kernel-selection mechanism; analogously, gradient ascent in the kernel parameter may be replaced by other suitable optimization schemes. Our implementation of the kernel update relies on the identity [6, 20]

$$\text{KSD}^2(\mu \mid \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k \, \mathrm{d}(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the required parameter gradient can be computed directly. Two observations are relevant for assessing the additional computational burden of Ad-SVGD relative to vanilla SVGD:

- The gradient-ascent updates for the kernel parameter reuse the same gradients $\nabla \log \pi(X_n^i)$ as the corresponding SVGD step, so Ad-SVGD does not require additional evaluations of $\nabla \log \pi$.
- The cost can be further controlled by using only a small number of kernel-parameter steps nstepstheta , or by updating the kernel parameter only after several SVGD iterations. For large particle counts M , subsampled particles may be used to approximate the KSD empirically.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $\text{nsteps}, \text{nstepstheta} \in \mathbb{N}$
Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to  $\text{nsteps} - 1$  do
    Decide  $\text{paramupdate} \in \{\text{True}, \text{False}\}$ ;
    if  $\text{paramupdate}$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $\text{nstepstheta} - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To motivate Ad-SVGD theoretically, we extend recent KSD-based convergence results for SVGD to the adaptive-kernel setting. More precisely, we adapt the analysis of [32], which is formulated under Talagrand’s inequality, a condition that holds under relatively mild assumptions on the target distribution and is weaker than the logarithmic Sobolev inequality often used in related work. The assumptions from [32] are imposed uniformly over the kernel-parameter family.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand's inequality T1, which means that there exists $\lambda > 0$ such that

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \| \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

In what follows, we use the notation

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and rely on the following fundamental inequality. For a fixed kernel parameter $\theta \in \Theta$ satisfying Assumption 3, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, take $\gamma, B > 0$, $\alpha > 1$, and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Proposition 3.1 of [32] then gives

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. Applied to the fixed-kernel iterates (3), this inequality yields the descent condition [32, Theorem 3.2]

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The essential verification behind (7) is the bound $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, obtained through Talagrand's inequality in Assumption 2, which permits the use of (6) with $g = \psi_\theta^{\mu_n}$. The descent condition (7) then implies

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

because $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) > 0$.

With an adaptive sequence of kernel parameters (θ_n) , the corresponding descent inequality becomes

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGD.

Suppose first that the adaptive SVGD iteration admits the idealized representation

$$\begin{aligned}\mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),\end{aligned}\tag{9}$$

which means that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu_n | \pi).$$

This formulation is available when the maximization of the KSD over the kernel parameter possesses a (unique) closed-form solution. Such a situation occurs, for example, in the multiple-kernel SVGD framework [1], where the kernel is parameterized as a convex combination of base kernels,

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

In the more general setting, however, the parameter update θ_n in (9) must be approximated. We assume throughout this idealized discussion that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The next descent statement follows directly from Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The analysis becomes more delicate when the KSD maximization over kernel parameters is solved only approximately. Assume that $\max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu | \pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned}\theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n\end{aligned}\tag{10}$$

for a sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ whose purpose is to increase $\operatorname{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is instantiated as gradient ascent on the KSD, although the same analysis allows other suitable iterative optimization schemes. The required property of the update rule is the following asymptotic accuracy condition.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) - \operatorname{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Under this assumption, we obtain the following convergence guarantee.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Applying Theorem 3.2 of [32] yields

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is monotonically decreasing and bounded below by zero, it converges and is therefore a Cauchy sequence. Consequently,

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Combining this fact with Assumption 4 gives

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under a Stein logarithmic Sobolev inequality, the preceding convergence statement can be strengthened to convergence in KL divergence. For several regimes governing the approximation errors ε_n , we derive iteration-complexity bounds for the mean-field limiting system (10); moreover, this framework explicitly characterizes the bias incurred when KSD maximization is solved only to a fixed $\bar{\varepsilon}$ -accuracy at each iteration. Details are provided in Section B. Finally, we stress that it is sufficient for only one kernel in the class $\{k_{\theta}\}$ to satisfy the Stein logarithmic Sobolev inequality, which leaves considerable flexibility for verifying the condition in practice.

5 Numerical Experiments

We evaluate Ad-SVGD on two primary numerical experiments: a one-dimensional Gaussian mixture model and a linear inverse problem governed by an ordinary differential equation. Additional controlled experiments probing the effect of increasing dimensionality are reported in Section A. In addition, Section 5.4 compares Ad-SVGD with SVGD using the median heuristic on a Bayesian logistic regression model previously used in the SVGD literature.

5.1 Kernel parameterization

SVGD is most often implemented with kernels of the form

$$k_h(x, y) = \exp \left(-\frac{\|x - y\|_p^p}{h} \right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. Our experiments focus on strategies for selecting the parameter h , the kernel *bandwidth*. The standard median heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, with med

denoting the current median inter-particle distance; this prescription is motivated by balancing the two terms in the SVGD update (3) [18].

To exploit the flexibility of the adaptive procedure, we use product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, thereby allowing bandwidths to vary by coordinate. The derivatives needed to apply Algorithm 1 with these kernels (see Equations (4) and (5)) are straightforward to compute. We also evaluated a coordinate-wise adaptation of the median heuristic that uses a naive median for each dimension, but this variant produced poor results and exhibited variance collapse.

Beyond the experiments reported here, the flexibility of Ad-SVGd permits more scalable kernel families in higher-dimensional settings. In particular, spectral or low-rank parameterizations can adapt dominant directions of variability without incurring the exponential cost of full product kernels, while Matérn or mixed-product constructions can provide additional control over smoothness and anisotropy.

5.2 Toy example

We first revisit the one-dimensional example of Liu and Wang [18], namely the two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We use $M \in \{50, 200, 500\}$ particles, initialize them from $\mathcal{N}(0, 1)$, and run SVGD for 10^4 iterations with step size 1, using kernels of the form described above with $p = 1$ and several fixed bandwidths h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit one-dimensional formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) and an exact sample of size 10^5 . The outcomes in Figure 1 show that accurate approximation occurs only within a restricted bandwidth range; consequently, the method is highly sensitive to h , making careful bandwidth selection essential.

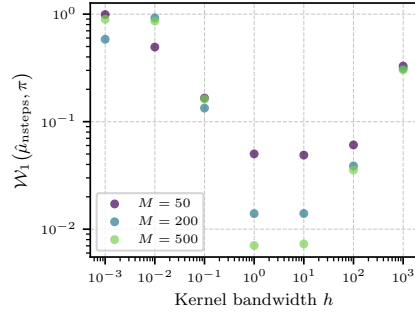


Figure 1: Final Wasserstein 1-distances in the one-dimensional example for SVGD with different fixed bandwidths h .

5.3 Linear inverse problem based on ODE

The next example is adapted from [38, Example 2.1]. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observations satisfy

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ denotes observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is given by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ evaluating the solution at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of this inverse problem, we impose a Gaussian process (GP) prior on u through the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and the coefficients $x_k \sim \mathcal{N}(0, \lambda_k)$ are independent with $\lambda_k = 50k^{-2}$. The inference task is therefore to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ under the Gaussian prior $\mathcal{N}(0, \Gamma_0)$, where $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$ yields the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For implementation, H is replaced by a numerical discretization operator for Equation (11) on a grid with mesh size 2^{-8} , and we consider the fully observed system with $N_{\text{obs}} = 2^8$. We retain $N_x = 16$ terms in the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare SVGD using the median heuristic, denoted Med-SVG, with Ad-SVG for particle ensemble sizes $M \in \{50, 100, 200\}$. Both methods use the kernels from Section 5.1 with $p = 1$ and are run for $4 \cdot 10^5$ iterations with particle-update step size 10^{-3} , together with the AdaGrad-style adaptive step-size control suggested by Liu and Wang [18]. For Ad-SVG, the bandwidth-update step size is 10^{-5} , and the bandwidth is updated only once every 100 particle updates; under this schedule, Med-SVG and Ad-SVG exhibited no significant runtime difference.

Figure 2 presents the GP reconstruction for a representative seed. Both methods approximate the posterior mean well, but only Ad-SVG captures the posterior uncertainty accurately. To quantify approximation quality, we compute the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance of the particle ensemble [29, Section 3.2]. Since the target π is multivariate normal, the distance admits the explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π denote the posterior mean and covariance. We also compare the marginal variances of the final particle distribution against those of the posterior. Figure 3 summarizes results aggregated over 56 random seeds; the posterior covariance, importantly, does not depend on $\bar{y}$. Consistently with the reconstruction, Ad-SVG yields more accurate posterior approximations than Med-SVG, which underestimates uncertainty. In addition, unlike Med-SVG, Ad-SVG improves in approximation quality as the particle count increases beyond 50. Figure 4 displays

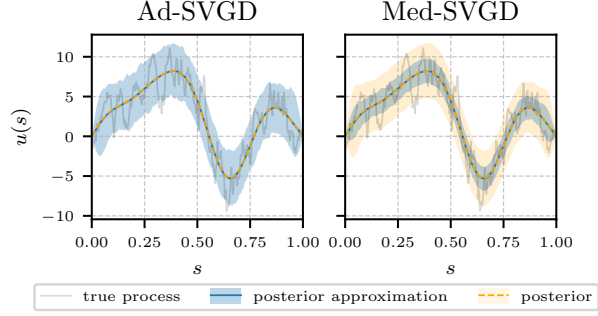
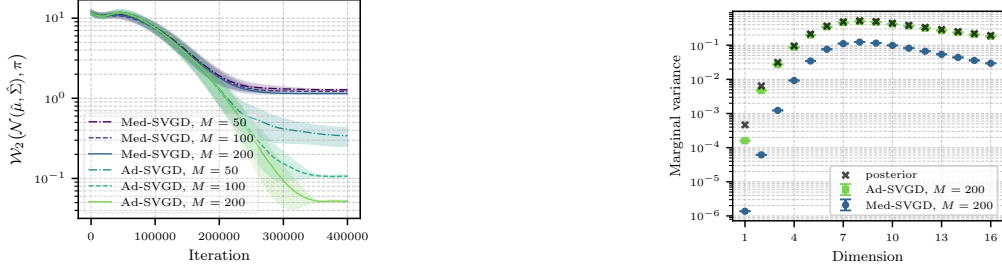


Figure 2: GP reconstruction for the ODE-based inverse problem, displaying the posterior mean and 90% confidence interval.

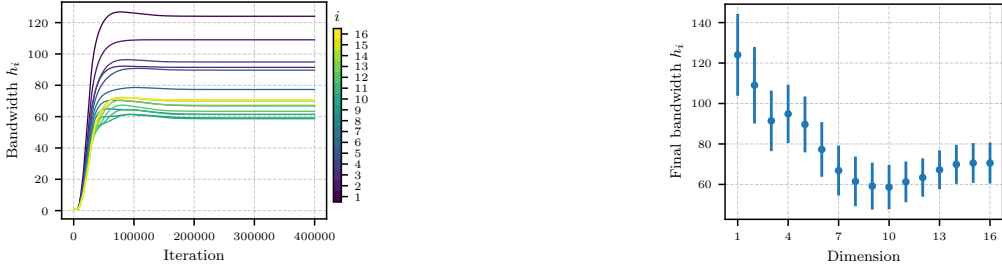
the bandwidths learned by Ad-SVGD, showing that the coordinate-wise bandwidths stabilize more rapidly than the approximation error and that the final bandwidths vary substantially across components, with a negative correlation to the corresponding marginal variances.



(a) Convergence profile of the approximation error.

(b) Estimated marginal variances.

Figure 3: Aggregated ODE-based inverse-problem results for Med-SVGD and Ad-SVGD, reported as mean and 95% confidence intervals over 56 random seeds.



(a) Mean evolution of the bandwidth in each dimension.

(b) Final bandwidths, shown as mean and 95% confidence interval.

Figure 4: Behavior of the bandwidth parameters for the ODE-based inverse problem under Ad-SVGD, aggregated over 56 random seeds.

5.4 Bayesian logistic regression

As in Liu and Wang [18] and Liu et al. [21], we consider Bayesian logistic regression (BLR) on the Covertypes data set [4]. For the regression weights w , we assign the Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, and we infer the posterior of $x = [w, \alpha]$. Following Liu et al. [21], we use the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] to obtain a reference against which the posterior approximation quality of Med-SVGD and Ad-SVGD is evaluated. Again following Liu et al. [21], we evaluate the methods on subsets of the original data set of size 1000 and aggregate results over 10 random draws. Classification performance is measured by test-set prediction accuracy using the particle mean. As shown in Figure 5, Ad-SVGD outperforms Med-SVGD for almost all seeds, achieving higher test accuracy while also remaining closer to the MCMC reference.

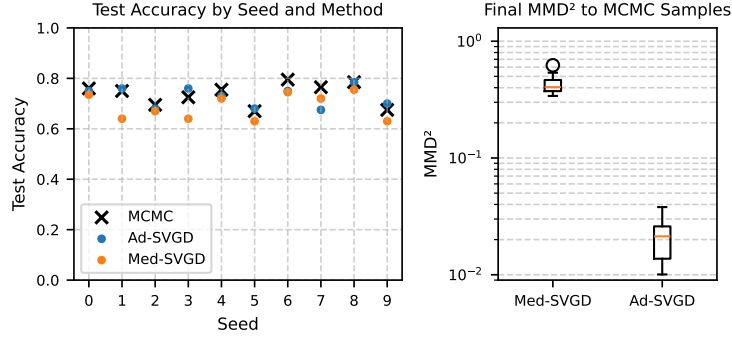


Figure 5: Approximation quality of SVGD particles for Bayesian logistic regression, measured by prediction accuracy (left) and MMD^2 to reference samples (right).

To assess posterior approximation accuracy, the same figure also reports the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] relative to the MCMC samples. The two methods differ markedly in their ability to capture the posterior distribution. This discrepancy is further corroborated by Figure 7, which compares covariance matrices of the final particle sets with the MCMC reference, averaged over the 10 seeds, using the visualization style of Liu et al. [21]. Med-SVG severely underestimates posterior uncertainty, whereas Ad-SVG provides a substantially closer approximation.

Moreover, Figure 6 illustrates the component-wise bandwidth trajectories: starting from the uninformative initialization of 1 in every component, Ad-SVG recovers suitable bandwidths that differ from the initialization by several orders of magnitude.

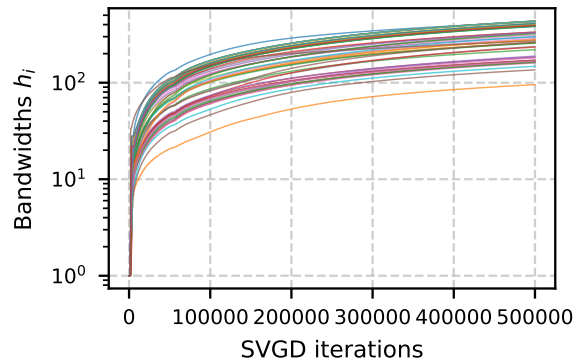


Figure 6: Bandwidth evolution in the BLR experiment.

6 Limitations and outlook

The principal limitation of our analysis is its reliance on Assumption 4. In ongoing work, we investigate when this condition is satisfied by the alternating gradient-ascent scheme in Algorithm 1 used throughout the experiments. Although the present paper does not provide a general verification of Assumption 4, the implemented procedure nevertheless yields promising empirical behavior.

Our analysis has deliberately focused on the original SVGD dynamics. A natural direction for future work is therefore to combine the proposed adaptive kernel-selection mechanism with more recent variants, including sliced SVGD [10], Grassmann SVGD [21], and Stein transport [27].

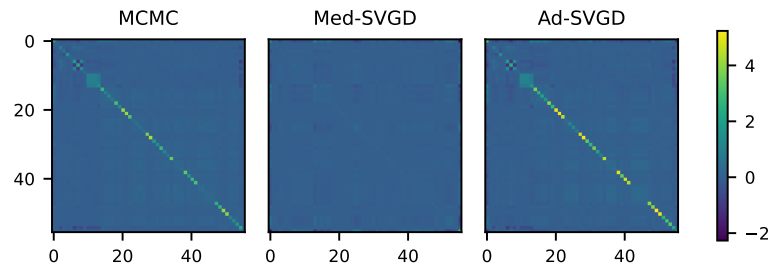


Figure 7: Covariance matrices of final SVGD particle sets compared with MCMC samples, averaged over 10 seeds.

References

- [1] Qingzhong Ai et al. “Stein Variational Gradient Descent with Multiple Kernels”. In: *Cognitive Computation* 15.2 (2023), pp. 672–682. DOI: [10.1007/s12559-022-10069-5](https://doi.org/10.1007/s12559-022-10069-5).
- [2] Jimmy Ba et al. “Understanding the Variance Collapse of SVGD in High Dimensions”. In: *International Conference on Learning Representations*. 2022.
- [3] Krishna Balasubramanian, Sayan Banerjee, and Promit Ghosal. “Improved Finite-Particle Convergence Rates for Stein Variational Gradient Descent”. In: *The Thirteenth International Conference on Learning Representations*. 2025.
- [4] Jock Blackard. *Coverttype*. UCI Machine Learning Repository. 1998. DOI: [10.24432/C50K5N](https://doi.org/10.24432/C50K5N).
- [5] Sinho Chewi et al. “SVGD as a kernelized Wasserstein gradient flow of the chi-squared divergence”. In: *Advances in Neural Information Processing Systems* 33. 2020.
- [6] Kacper Chwialkowski, Heiko Strathmann, and Arthur Gretton. “A Kernel Test of Goodness of Fit”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [7] Gianluca Detommaso et al. “A Stein variational Newton method”. In: *Advances in Neural Information Processing Systems* 31. 2018.
- [8] Andrew Duncan, Nikolas Nüsken, and Lukasz Szpruch. “On the geometry of Stein variational gradient descent”. In: *Journal of Machine Learning Research* 24 (2023).
- [9] Rémi Flamary et al. “POT: Python Optimal Transport”. In: *Journal of Machine Learning Research* 22 (2021).
- [10] Wenbo Gong, Yingzhen Li, and José Miguel Hernández-Lobato. “Sliced Kernelized Stein Discrepancy”. In: *International Conference on Learning Representations*. 2021.
- [11] Jackson Gorham and Lester Mackey. “Measuring Sample Quality with Kernels”. In: *Proceedings of the 34th International Conference on Machine Learning*. 2017.
- [12] Arthur Gretton et al. “A Kernel Two-Sample Test”. In: *Journal of Machine Learning Research* 13 (2012).
- [13] Matthew D. Hoffman and Andrew Gelman. “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo”. In: *Journal of Machine Learning Research* 15 (2014).
- [14] Richard Jordan, David Kinderlehrer, and Felix Otto. “The Variational Formulation of the Fokker–Planck Equation”. In: *SIAM Journal on Mathematical Analysis* 29.1 (1998), pp. 1–17. DOI: [10.1137/S0036141096303359](https://doi.org/10.1137/S0036141096303359).
- [15] Rahif Kassab and Osvaldo Simeone. “Federated Generalized Bayesian Learning via Distributed Stein Variational Gradient Descent”. In: *IEEE Transactions on Signal Processing* 70 (2022), pp. 2180–2192. DOI: [10.1109/TSP.2022.3168490](https://doi.org/10.1109/TSP.2022.3168490).
- [16] Anna Korba et al. “A Non-Asymptotic Analysis for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems* 33. Vol. 33. 2020.
- [17] Lauro Langosco di Langosco, Vincent Fortuin, and Heiko Strathmann. “Neural Variational Gradient Descent”. In: *Fourth Symposium on Advances in Approximate Bayesian Inference*. 2022.

- [18] Q. Liu and D. Wang. “Stein Variational Gradient Descent: A General Purpose Bayesian Inference Algorithm”. In: *Advances in Neural Information Processing Systems 29*. 2016.
- [19] Qiang Liu. “Stein Variational Gradient Descent as Gradient Flow”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [20] Qiang Liu, Jason Lee, and Michael Jordan. “A Kernelized Stein Discrepancy for Goodness-of-fit Tests”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [21] Xing Liu et al. “Grassmann Stein Variational Gradient Descent”. In: *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*. 2022.
- [22] Yang Liu et al. “Stein Variational Policy Gradient”. In: *Proceedings of the Thirty-Third Conference on Uncertainty in Artificial Intelligence*. 2017.
- [23] Jianfeng Lu, Yulong Lu, and James Nolen. “Scaling Limit of the Stein Variational Gradient Descent: The Mean Field Regime”. In: *SIAM Journal on Mathematical Analysis* 51.2 (2019), pp. 648–671. DOI: [10.1137/18M1187611](https://doi.org/10.1137/18M1187611).
- [24] Safa Messaoud et al. “S2AC: Energy-Based Reinforcement Learning with Stein Soft Actor Critic”. In: *The Twelfth International Conference on Learning Representations*. 2024.
- [25] Krikamol Muandet et al. “Kernel Mean Embedding of Distributions: A Review and Beyond”. In: *Foundations and Trends in Machine Learning* 10.1-2 (2017), pp. 1–141. DOI: [10.1561/22000000060](https://doi.org/10.1561/22000000060).
- [26] Alfred Müller. “Integral Probability Metrics and Their Generating Classes of Functions”. In: *Advances in Applied Probability* 29.2 (1997), pp. 429–443. DOI: [10.2307/1428011](https://doi.org/10.2307/1428011).
- [27] Nikolas Nüsken. *Stein transport for Bayesian inference*. 2024. arXiv: [2409.01464](https://arxiv.org/abs/2409.01464).
- [28] Nikolas Nüsken and D. R. Michiel Renger. “Stein variational gradient descent: Many-particle and long-time asymptotics”. In: *Foundations of Data Science* 5.3 (2023), pp. 286–320. DOI: [10.3934/fods.2022023](https://doi.org/10.3934/fods.2022023).
- [29] Victor M. Panaretos and Yoav Zemel. “Statistical Aspects of Wasserstein Distances”. In: *Annual Review of Statistics and Its Application* 6 (2019), pp. 405–431. DOI: [10.1146/annurev-statistics-030718-104938](https://doi.org/10.1146/annurev-statistics-030718-104938).
- [30] Yuchen Pu et al. “VAE Learning via Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [31] Sebastian Reich and Simon Weissmann. “Fokker–Planck Particle Systems for Bayesian Inference: Computational Approaches”. In: *SIAM/ASA Journal on Uncertainty Quantification* 9.2 (2021), pp. 446–482. DOI: [10.1137/19M1303162](https://doi.org/10.1137/19M1303162).
- [32] Adil Salim, Lukang Sun, and Peter Richtarik. “A Convergence Theory for SVGD in the Population Limit under Talagrand’s Inequality T1”. In: *Proceedings of the 39th International Conference on Machine Learning*. 2022.
- [33] Jiaxin Shi and Lester Mackey. “A Finite-Particle Convergence Rate for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 36*. 2023.
- [34] Björn Sprungk, Simon Weissmann, and Jakob Zech. “Metropolis-adjusted interacting particle sampling”. In: *Statistics and Computing* 35.64 (2025). DOI: [10.1007/s11222-025-10595-w](https://doi.org/10.1007/s11222-025-10595-w).

- [35] Charles Stein. “A bound for the error in the normal approximation to the distribution of a sum of dependent random variables”. In: *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Volume 2: Probability Theory*. 1972, pp. 583–603.
- [36] Dilin Wang, Zhe Zeng, and Qiang Liu. “Stein Variational Message Passing for Continuous Graphical Models”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.
- [37] Dilin Wang et al. “Stein Variational Gradient Descent With Matrix-Valued Kernels”. In: *Advances in Neural Information Processing Systems 32*. 2019.
- [38] Simon Weissmann, Ashia Wilson, and Jakob Zech. “Multilevel Optimization for Inverse Problems”. In: *Proceedings of Thirty Fifth Conference on Learning Theory*. 2022.
- [39] Qian Zhao et al. “Stein variational gradient descent with learned direction”. In: *Information Sciences* 637 (2023). DOI: [10.1016/j.ins.2023.118975](https://doi.org/10.1016/j.ins.2023.118975).
- [40] Jingwei Zhuo et al. “Message Passing Stein Variational Gradient Descent”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.

A More numerical experiments

A.1 Gaussian mixture models

We revisit the example from Section 5.2. For the setup specified there, Figure 9 displays histograms of the final particle distributions against the target density for $h \in \{0.001, 1, 1000\}$. We then compare Med-SVGD and Ad-SVGD across several particle counts M , evaluating the final particle distributions produced after 10^4 iterations with step size 1.

Figure 8 reports the Wasserstein 1-distance between the final empirical particle law and the target distribution for Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. As expected, approximation quality improves with N for both algorithms, and the two methods achieve comparable performance, reaching Wasserstein distance below 0.01 for $N = 500$.

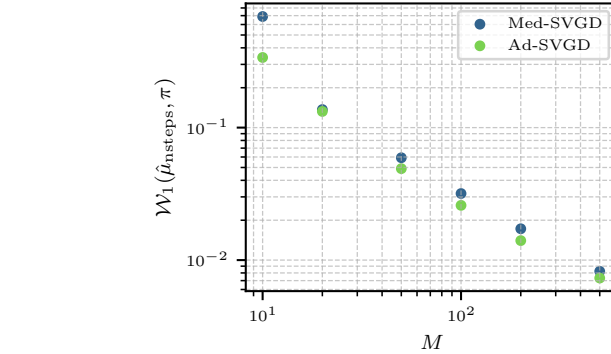


Figure 8: Approximation error for the one-dimensional Gaussian mixture.

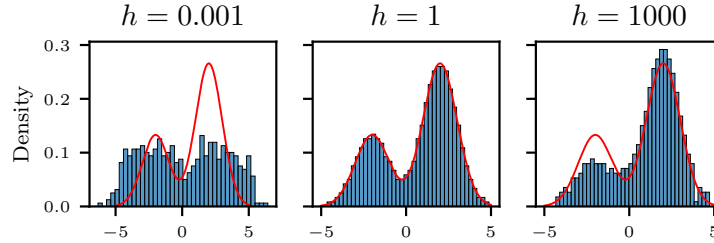


Figure 9: Histograms of final particle sets generated with fixed bandwidths $h = 0.001, 1, 1000$; the target density π is shown in red for comparison.

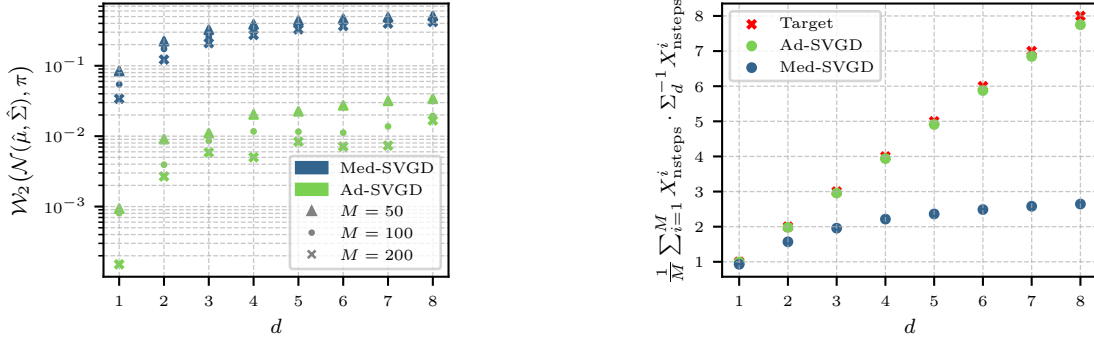
In the one-dimensional setting, both methods approximate the target distribution accurately. The adaptive bandwidth-selection strategy is effective, but in this simple regime it does not confer a substantial advantage over the commonly used median heuristic.

A.2 Scaling dimension: multivariate normal distribution

Turning to higher-dimensional targets, we consider Gaussian distributions $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. The initial particle distribution is $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$, and the algorithms are run for 10^4 iterations with step size 0.1. When numerical stability required it, we used a smaller step size and adjusted the number of iterations accordingly.

We compare the approximation quality of the final particle ensembles generated by Med-SVGD and Ad-SVGD as the dimensionality increases. Since the Wasserstein 2-distance between Gaussian distributions has an explicit formula (see [29]), sample quality is measured by the Wasserstein

2-distance between the target π_d and the Gaussian distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance of the particles [29, Section 3.2]. Following the terminology used by the corresponding function in the Python Optimal Transport library [9], which we use for the computations, we refer to this quantity as the *Bures Wasserstein distance*. Figure 10a shows the evolution of this measure for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as dimension increases, demonstrating that Ad-SVGD substantially outperforms Med-SVGD for all considered d .



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal target distributions of increasing dimension.

Because the Bures Wasserstein distance is relatively insensitive to the smaller variances in the final coordinates of the targets π_d , we additionally examine the statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom, and hence has expectation d , when $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). We compute the empirical mean of this statistic over the particle set with $M = 200$ and compare it with the expectation d in Figure 10b. Beyond the one-dimensional case, the Med-SVGD statistic increasingly departs from the correct expectation as d grows, whereas the Ad-SVGD statistic remains comparatively close, indicating that Med-SVGD fails to represent higher-dimensional uncertainty accurately while Ad-SVGD remains effective.

Further details on the final particle distributions for $M = 200$ and varying d are provided in Tables 1 and 2. For each d , these tables compare the marginal particle variances with the corresponding target variances, which are displayed in the first row. Table 1 reports Med-SVGD, whereas Table 2 reports Ad-SVGD. The adaptive method approximates the target variance well across all components, while Med-SVGD substantially underestimates the target uncertainty, consistent with the diagnostic behavior in Figure 10b.

Finally, we examine the marginal particle distributions for $M = 200$ and $d = 8$. To facilitate visualization, each coordinate is rescaled by the inverse of its target marginal standard deviation, equivalently multiplying the i -th component by i , so that each marginal of π_d becomes standard normal. Figures 11 and 12 show histograms of these normalized marginals for particles generated by Med-SVGD and Ad-SVGD, respectively, with the standard Gaussian density included for comparison. Again, Ad-SVGD captures all marginals well, whereas Med-SVGD underestimates target uncertainty. The same conclusion is visible in Figures 13 and 14, where normalized marginal particle quantiles are compared with standard normal target quantiles.

Table 1: Marginal variances of the final particle distributions in the d -dimensional examples generated by Med-SVGD with $M = 200$; target marginal variances for π_d are included for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final particle distributions in the d -dimensional examples generated by Ad-SVGD with $M = 200$; target marginal variances for π_d are included for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

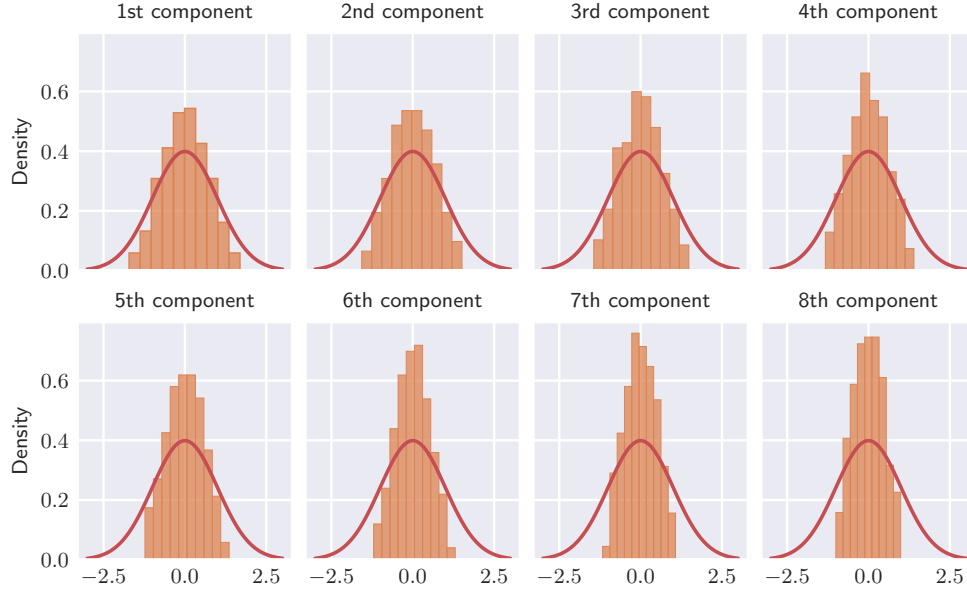


Figure 11: Histograms of individual components of the final particle set for the eight-dimensional example generated by Med-SVGD with $M = 200$, rescaled using target marginal variances; the standard Gaussian density is shown for comparison.

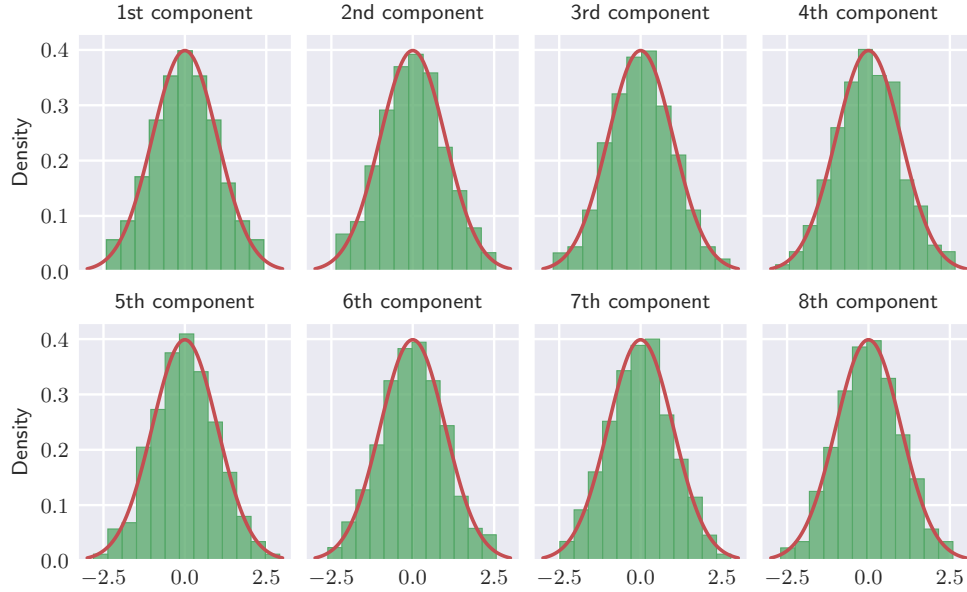


Figure 12: Histograms of individual components of the final particle set for the eight-dimensional example generated by Ad-SVGD with $M = 200$, rescaled using target marginal variances; the standard Gaussian density is shown for comparison.

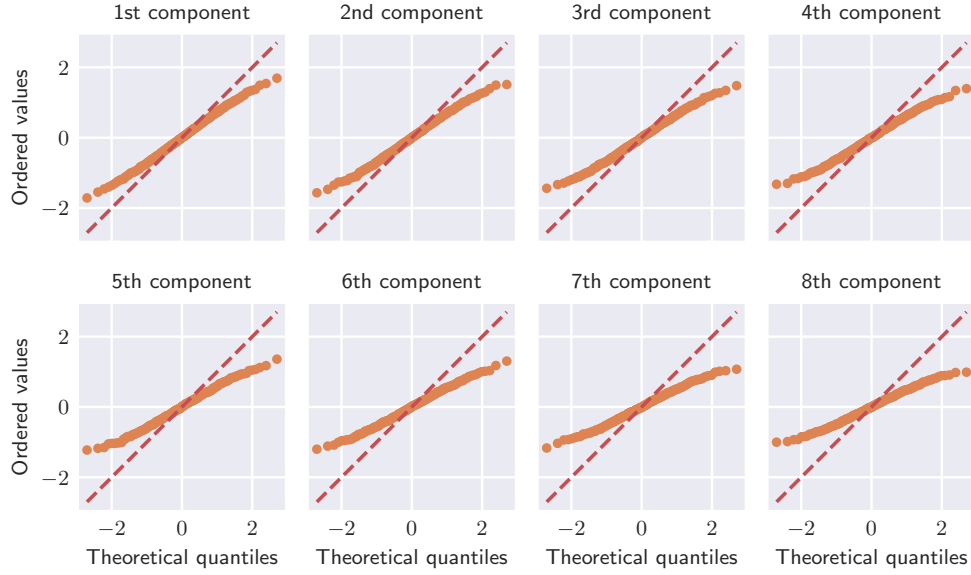


Figure 13: Q-Q plots comparing rescaled marginals of the final particle distribution for the eight-dimensional Med-SVGd example with $M = 200$ against a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

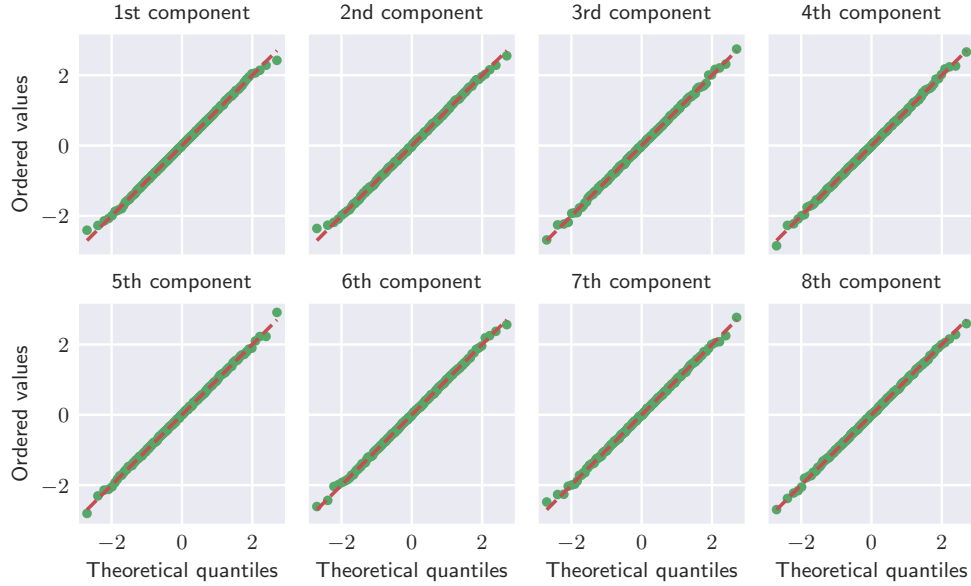


Figure 14: Q-Q plots comparing rescaled marginals of the final particle distribution for the eight-dimensional Ad-SVGd example with $M = 200$ against a standard normal distribution; the line of slope 1 through the origin is shown for comparison.

A.3 Gaussian process inference

We further examine the dimension-scaling behavior of Ad-SVGD in the Gaussian process (GP) inference task proposed in [31]. Consider a GP on $[0, 1]$ represented by the truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ consists of independent Gaussian variables $x_k \sim \mathcal{N}(0, k^{-2})$. The process is observed at N_y equispaced points in $[0, 1]$, and the goal is to infer the coefficients x_k . For fixed N_x and N_y , this is an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$ and $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$, with diagonal covariance entries k^{-2} for $k = 1, \dots, N_x$, and the observational noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

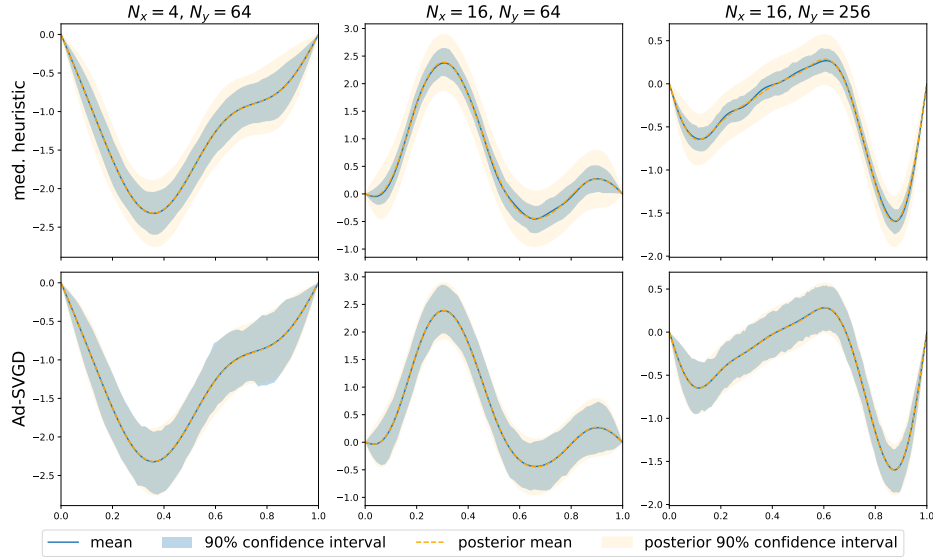


Figure 15: Estimated processes generated by the median heuristic and Ad-SVGD with $M = 100$ particles for different model configurations, compared with the posterior.

We use SVGD to sample from the posterior $X | Y = \bar{y}$ and compare the median heuristic with the adaptive strategy. For SVGD with the median heuristic, the kernel is $k(x, y) = \exp(-\|x - y\|_1/h)$; for Ad-SVGD, we use the product kernel $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, assigning a separate bandwidth to each coordinate. Following [18], we use a variant of Adagrad for step-size control when $N_x = 16$.

Figure 15 compares the inferred processes with the posterior for $N_x = 4, N_y = 64$; $N_x = 16, N_y = 64$; and $N_x = 16, N_y = 256$. As dimensionality increases, SVGD with the median heuristic underestimates posterior variance, whereas Ad-SVGD provides a closer approximation, and this behavior persists across different numbers of GP observations. In Table 3, we quantify the effect by comparing the trace of the covariance of the SVGD particle distributions with the true posterior trace. The median heuristic substantially underestimates uncertainty, while Ad-SVGD captures the variance more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the covariance of the final particle distribution ($M = 100$ particles) compared with the theoretical posterior across configurations, averaged over 25 runs.

B Refined theoretical analysis

The convergence result can be sharpened under a Stein log-Sobolev inequality, which relates KL divergence to the KSD. This inequality was introduced by Duncan, Nüsken, and Szpruch [8], who also discussed conditions under which it may hold. In the present adaptive kernel-selection setting, the requirement can be relaxed in a useful way: it is enough that a single kernel within the entire kernel class satisfies the log-Sobolev inequality. More precisely, we impose the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property allows convergence to be quantified directly in KL divergence. Notably, in this regime we can bound $\mathcal{F}(\mu_n)$ even when the approximation error satisfies only the uniform condition $\varepsilon_n \leq \bar{\varepsilon}$, without requiring $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}$$

for all $n \in \mathbb{N}$.

Proof. As in the proof of Theorem 3, Assumption 5 together with the stated conditions on α, γ gives

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iteration of this recursion over $n \in \mathbb{N}$ yields the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We now treat the three cases separately:

1. If $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows from the geometric-series estimate

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1-\rho}.$$

2. If $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1-\rho/q} = q^n \frac{1}{q-\rho}$$

using the geometric-series formula. Similarly, when $q < \rho$, we directly obtain

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1-q/\rho} = \rho^n \frac{1}{(\rho-q)}.$$

Finally, when $q = \rho$, the bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ permits the same line-by-line argument.

3. Under the polynomial decay condition $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here we used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□